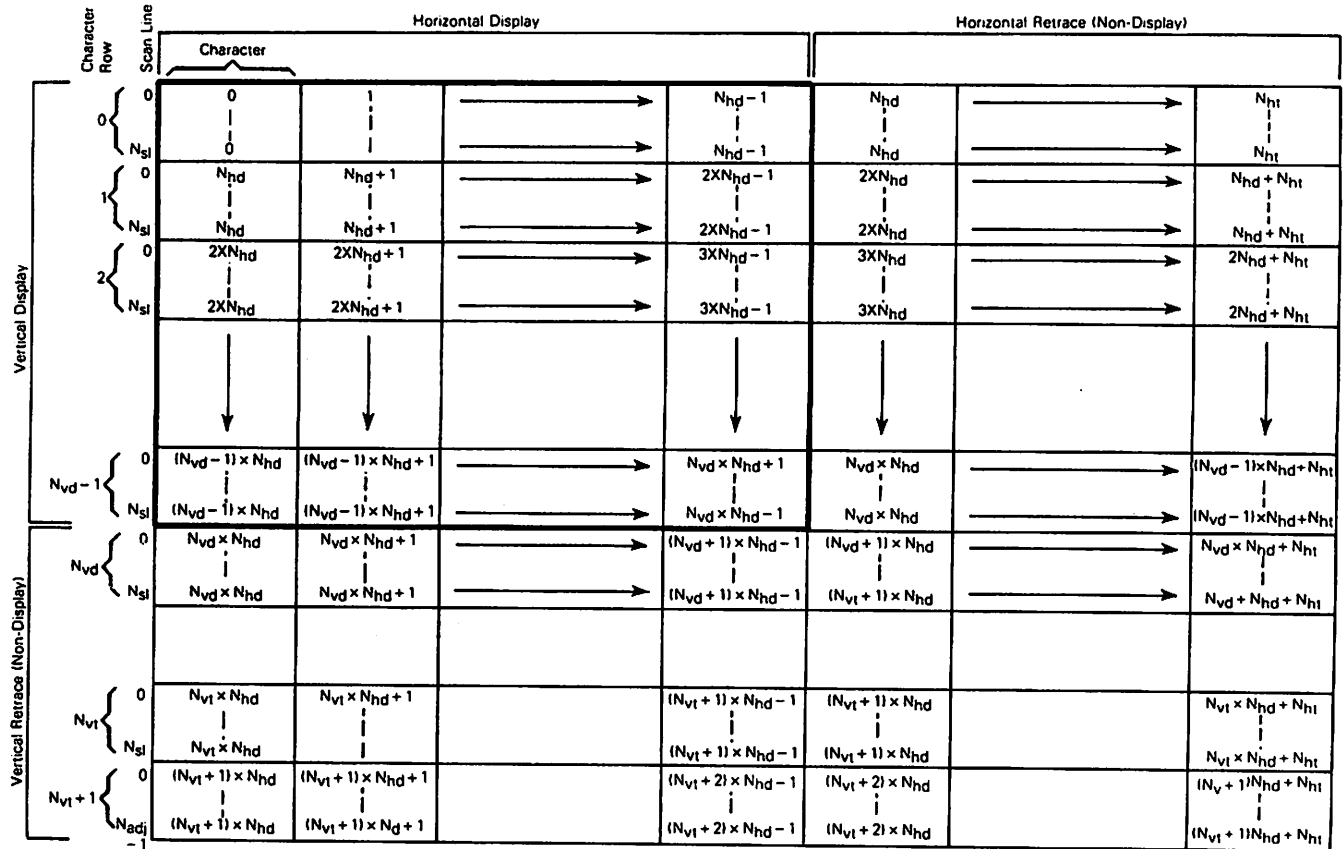


FIGURE 18 — REFRESH MEMORY ADDRESSING (MA0-MA13) STAGE CHART



NOTE 1: The initial MA is determined by the contents of start address register, R12/R13. Timing is shown for R12/R13=0. Only non-interface and interface sync modes are shown.

6845 Programming Info.

DETERMINING REGISTER CONTENTS

Some of the register contents are determined rather easily. They are:

Register	Name	Contents
R8	Interlace Mode Register	See Table 3
R10	Cursor Start	See Figure 15 and Table 4
R11	Cursor End	See Figure 15
R12	Start Address (H)	User programs first memory location to be displayed
R13	Start Address (L)	User programs desired cursor location
R14	Cursor (H)	Can be loaded via light-pen stroke only
R15	Cursor (L)	
R16	Light Pen (H)	
R17	Light Pen (L)	

The remaining register contents must be determined from some basic data related to the CRT monitor and from the user-desired display format. The CRTC reference sheet (see Figure 19) gives a set of formulas for calculating the register contents as well as other useful characteristics of the display. This type of data is summarized under basic parameters in Figures 20 and 21; most or all of this data must be supplied by the user before he can determine the contents for registers R0-R7 and R9. All variables B1-B10 are equal to basic parameters 1 through 10.

FIGURE 19 - CRTC REFERENCE SHEET

Register Function		Intermediate Calculations			Register Calculations	
		Symbol	Description	Calculation	Register	Calculation
R0	Horizontal Total					
R1	Horizontal Displayed	f'	Dot frequency (1st approx.)	$\frac{B_5 \cdot (B_7 + B_9)}{(1/B_1) - B_3}$	R0	$\frac{f'}{B_1 \cdot (B_7 + B_9)} - 1$
R2	Horizontal Sync Position					
R3	Horizontal Sync Width	t_c	Character Time	$\frac{1}{[(R0) + 1] \cdot B_1}$	R1	B_5
R4	Vertical Total	f	Dot frequency	$\frac{B_7 + B_9}{t_c}$		
R5	Vertical Total Adjust				R2	$(R1) + \frac{(R3)}{2}$
R6	Vertical Displayed	t_{sl}	Scan line time	$[(R0) + 1] \cdot t_c$	R3	$\frac{(R0) - (R1)}{3}$
R7	Vertical Sync Position	n	Total # of scan lines	$\frac{1}{B_2 \cdot t_{sl}}$		
R8	Interlace Mode				R4	$N - 1$
R9	Maximum Scan Line Address	N	Integer and	$\frac{n}{B_8 + B_{10}} = \frac{N + R}{B_8 + B_{10}}$	R5	R
R10	Cursor Start	R	Integer remainder			
R11	Cursor End				R6	B_6
R12	Start Address (H)	t_{cr}	Character row time	$(B_8 + B_{10}) \cdot t_{sl}$	R7	$[(R4) + 1] - \frac{16 - (R5) \geq (R7) \geq (R6)}{B_8 + B_{10}}$
R13	Start Address (L)	t_{hr}	Horizontal retrace time	$\leq \frac{[(R0) + 1 - B_5] \cdot (B_7 + B_9)}{f}$		
R14	Cursor (H)				R9	$(B_8 + B_{10}) - 1$
R15	Cursor (L)	t_{vr}	Vertical retrace time	$\leq \frac{B_1}{B_2} - B_6(B_8 + B_{10}) \cdot t_{sl}$		
R16	Light Pen (H)					
R17	Light Pen (L)					

In Figures 20 and 21, worksheet example calculations are shown for 32 x 16 and 80 x 24 display formats respectively. The following items are keyed to the figures. Basic parameters ① through ⑩ have been provided; items ① through ④ are data about the CRT monitor and items ⑤ through ⑩ are data about the user's desired display.

- ① Calculate the approximate dot frequency. The user should verify that the bandwidth of his CRT monitor will accommodate this frequency.
- ② Calculate R0. The resultant answer will usually be an integer plus a fraction. Assume the next high integer.
- ③ Fill in value for R1.
- ④ Calculate R3. Use the next highest integer. In these examples the sync width was chosen to be one third of the horizontal blanking interval.
- ⑤ Calculate R2. Again, use the next highest integer.
- ⑥ Calculate t_c , character tie. This is the time required for one scan line of one character block to be written.
- ⑦ Calculate the exact dot frequency.
- ⑧ Calculate t_{sl} , scan line time. This is the time required for one scan line of one character row to be written including retrace time.

- ⑨ Calculate n. This is the total number of scan lines for each frame. Discard any fraction.
- ⑩ Calculate N and R.
- ⑪ Calculate R4.
- ⑫ Fill in R5.
- ⑬ Fill in R6.
- ⑭ Calculate R7. If there is no possible value for R7, then the display demands for the CRT monitor exceed its capability. A compromise adjustment must be made in basic parameter 6, 8, or 10.
- ⑮ Calculate R9.
- ⑯ Calculate t_{cr} . This the time required for one character row to be written.
- ⑰ Calculate t_{hr} . $t_{hr} > B_3$.
- ⑱ Calculate t_{vr} . $t_{vr} > B_4$.

In Figure 20, calculation ⑧ verifies that the vertical period is 16.667 milliseconds or 60 hertz. The expression used is:

$$t_{cr} \times [(R4) + 1] + [t_{sl} \times (R5)] = Vp.$$

Another check is calculation of horizontal sync pulse width R3. $t_c = PW_{HS}$ (typically approximately equals 4 microseconds).

For convenience, a blank worksheet is provided in Figure 22.

FIGURE 20 — CRT WORKSHEET EXAMPLE CALCULATION (32 x 16)

Basic Parameters (B1-B10)		Intermediate Calculations		Register Calculations			
	Symbol		Value	Register	Decimal	Hex	
1. Horizontal frequency	$= \frac{15750 \times 500}{15750}$ ① f'	$\frac{32 \times (5+2)}{1 - 11 \times 10^{-6}}$	$= 4.27 \times 10^6$	② R0	$\frac{4.27 \times 10^6}{15,750 \times (5+2) - 1}$	38	26
2. Vertical frequency	$= \frac{60}{39 \times 15750}$ ⑥ t_c	$\frac{1}{39 \times 15750}$	$= 1.63 \times 10^{-6}$	③ R1	$B_5 = 32$	32	20
3. Minimum Horizontal retrace time	$= \frac{11 \times 10^{-6}}{1.63 \times 10^{-6}}$ ⑦ t	$\frac{5+2}{1.63 \times 10^{-6}}$	$= 4.29 \times 10^6$	⑤ R2	$32 + \frac{3}{2}$	33	21
4. Minimum vertical retrace time	$= \frac{10^{-3}}{39 \times 1.63 \times 10^{-6}}$ ⑧ t_{sl}	$39 \times 1.63 \times 10^{-6}$	$= 63.6 \times 10^{-6}$	④ R3	$\frac{38 - 32}{3}$	2	2
5. # of displayed characters per row	$= \frac{32}{60 \times 63.6 \times 10^{-6}}$ ⑨ n	$\frac{1}{60 \times 63.6 \times 10^{-6}}$	$= 262$	⑪ R4	$17 - 1 =$	16	10
6. # of displayed character rows	$= \frac{16}{7+8}$ ⑩ N	$\frac{262}{7+8}$	$= 17$	⑫ R5	$R = 7$	7	7
7. # of dots in character dot matrix row	$= \frac{5}{7+8}$ R	$\frac{262}{7+8}$	$= 7$	⑬ R6	$B_6 = 16$	16	10
8. # of scan lines in character * matrix column	$= \frac{7}{16(7+8) \times 63.6 \times 10^{-6}}$ ⑬ t_{cr}	$(7+8) \times 63.6 \times 10^{-6}$	$= 954 \times 10^{-6}$	⑭ R7	(A)	16	10
9. Number of dots between horizontal adjacents	$= \frac{2}{4.29 \times 10^6}$ ⑰ t_{hr}	$\frac{(38+1-32) \times (5+2)}{4.29 \times 10^6}$	$= 11.42 \times 10^{-6}$	R8			
10. Number of scan lines between vertical adjacents	$= \frac{8}{16(7+8) \times 63.6 \times 10^{-6}}$ ⑱ t_{vr}	$\frac{(15750 - 16(7+8)) \times 63.6 \times 10^{-6}}{60}$	$= 1.431 \times 10^{-3}$	⑮ R9	$7+8-1 =$	14	0E
① $16 + 1 - \frac{16-7}{7+8} \geq (R7) \geq 16$				R10			
② $17 - 6 \geq (R7) \geq 16$				R11			
③ $954 \times 10^{-6} \times 17 = 16.218 \times 10^{-3}$				R12			
④ $+ \frac{63.6 \times 10^{-6} \times 7 = 445 \times 10^{-3}}{16.663 \text{ ms}} = \frac{1}{1}$				R13			
⑤ $f = 60 \text{ Hz}$				R14			
				R15			

FIGURE 21 — CRTC WORKSHEET EXAMPLE CALCULATION (80 x 24)

Basic Parameters (B1-B10)		Intermediate Calculations		Register Calculations			
	Symbol		Value	Register	Decimal	Hex	
1. Horizontal frequency	= <u>18,600</u> (1) f'	$\frac{80 \cdot (7+2)}{18600 - 11 \times 10^{-6}}$	$16 \cdot 836 \times 10^6$	(2) R0	$\frac{16.836 \times 10^6 - 1}{(18,600)(9)}$	100	64
2. Vertical frequency	= <u>60</u> (6) f _c	$\frac{1}{(100+1) \cdot 18600}$	532.31×10^{-9}	(3) R1	85 = 80	80	50
3. Minimum Horizontal retrace time	= 11×10^{-6} (7) f	$\frac{7+2}{532.31 \times 10^{-9}}$	16.907×10^6	(5) R2	$80 + \frac{7}{2}$	84	54
4. Minimum vertical retrace time	= 1×10^{-3} (8) t _{sl}	$(100+1)(532.31 \times 10^{-9})$	53.76×10^{-6}	(4) R3	$\frac{R0 - R1}{3}$	7	07
5. # of displayed characters per row	= <u>80</u> (9) n	$\frac{1}{(60)(53.76 \times 10^{-6})}$	310	(11) R4	28 - 1	27	18
6. # of displayed character rows	= <u>24</u> (10) N		28	(12) R5	R = 2	02	02
7. # of dots in character dot matrix row	= <u>7</u> R	$\frac{310}{11}$	2	(13) R6	86 = 24	24	18
8. # of scan lines in character • matrix column	= <u>9</u> (16) t _{cr}	$(9+2)(53.76 \times 10^{-6})$	591.39×10^{-6}	(14) R7	(A)	25	19
9. Number of dots between horizontal adjacents	= <u>2</u> (17) t _{hr}	$\leq \frac{(101-80)(7+2)}{16.907 \times 10^6}$	11.17×10^{-6}	(15) R9	$(9+2) - 1$	10	0A
10. Number of scan lines between vertical adjacents	= <u>2</u> (18) t _{vr}	$\leq \left[\frac{18600}{60} - 24(11) \right] 53.76 \times 10^{-6}$	2.47×10^{-3}	R10		00	00
(A) $\frac{(27+1) - (16-2)}{11} \geq R7 \geq 24$ 26 72 ≥ R7 ≥ 24		(B) B2 = $\frac{1}{1/(t_{cr})(R4+1) + (t_{sl})(R5)}$ = $\frac{1}{1/(591.39 \times 10^{-6})(28) + (53.76 \times 10^{-6})(2)}$ = $\frac{1}{16.667 \times 10^{-3}}$ = 60		R11	11	0B	
				R12		00	
				R13	128	80	
				R14	128	00	
				R15		80	

FIGURE 22 — CRTC WORKSHEET

Basic Parameters		Intermediate Calculations		Register Calculations			
		Symbol	Value	Register	Decimal	Hex	
1. Horizontal frequency	=	f'	_____	R0	_____	_____	
2. Vertical frequency	=	f_c	_____	R1	_____	_____	
3. Minimum Horizontal retrace time	=	t	_____	R2	_____	_____	
			_____	R3	_____	_____	
			_____	R4	_____	_____	
4. Minimum vertical retrace time	=	t_{sl}	_____	R5	_____	_____	
			_____	R6	_____	_____	
5. # of displayed characters per row	=	n	_____	R7	_____	_____	
			_____	R8	_____	_____	
6. # of displayed character rows	=	N	_____	R9	_____	_____	
			_____	R10	_____	_____	
7. # of dots in character dot matrix row	=	R	_____	R11	_____	_____	
			_____	R12	_____	_____	
8. # of scan lines in character * matrix column	=	t_{cr}	_____	R13	_____	_____	
			_____	R14	_____	_____	
9. Number of dots between horizontal adjacents	=	t_{hr}	_____	R15	_____	_____	
			_____	R16	_____	_____	
10. Number of scan lines between vertical adjacents	=	t_{vr}	_____	R17	_____	_____	
			_____	R18	_____	_____	
				R19	_____	_____	

CRTC INITIALIZATION

Register R0-R15 must be initialized after the system is powered up. The processor will normally load the CRTC register file from a firmware table. The program required to initialize the CRTC for a 80x24 format (example calculation #2) is shown in Figure 23.

The CRTC registers will have an initial value at power up. When using a direct drive monitor (sans horizontal oscillator) these initial values may result in out-of-tolerance operation. CRTC programming should be done immediately after power up especially in this type of system.

ADDITIONAL CRTC APPLICATIONS

The foremost system function which may be performed by the CRTC controller is the refreshing of dynamic RAM. This

is quite simple as the refresh addresses continually run.

Note that the LPSTB input may be used to support additional system functions other than a light pen. A digital-to-analog converter (DAC) and comparator could be configured to use the refresh addresses as a reference to a DAC composed of a resistive adder network connected to a comparator. The output of the comparator would generate the LPSTB input signifying a match between the refresh address analog level and the unknown voltage.

The light-pen strobe input could also be used as a character strobe to allow the CRTC refresh addresses to decode a keyboard matrix. Debouncing would need to be done in software.

Both the VS and HS outputs may be used as a real-time clock. Once programmed, the CRTC will provide a stable reference frequency.

PAGE 001 CRTCINIT.SA:0 MC6845 CRTC Initialization Program

```

00001          NAM      MC6845
00002          TTL      / MC6845-1 CRTC initialization program
00003          OPT      G,S,LLE=85 print FCB's, FDB's & XREF table
00004          *****
00005          * Assign CRTC addresses
00006          *
00007          9000 A CRTCAD EQU      $9000      Address Register
00008          9001 A CRTCRG EQU      CRTCAD+1  Data Register
00009          *****
00010          * Initialization program
00011          *
00012A 0000          ORG      0              a place to start
00013A 0000 5F          CLRB              clear counter
00014A 0001 CB 1020 A   LDX      #CRTTAB    table pointer
00015A 0004 F7 9000 A   STAB     CRTCAD     load address register
00016A 0007 A6 00 A     LDAA     0,X        get register value from table
00017A 0009 B7 9001 A   STAA     CRTCRG     program register
00018A 000C 08          INX              increment counters
00019A 000D 5C          INCB
00020A 000E C1 10 A     CMPB     $10        finished?
00021A 0010 26 F2 0004 BNE     CRTC1       no: take branch
00022A 0012 3F          SWI              yes: call monitor
00023          *****
00024          * CRTC register initialization table
00025          * 80 x 24 non-interlaced format
00026A 1020          ORG      $1020        start of table
00027A 1020 65 A CRTTAB FCB $64,$50 R0, R1 - H total & H displayed
00028A 1022 56 A          FCB     $54,$07  R2, R3 - HS pos. & HS width
00029A 1024 18 A          FCB     $1B,$02  R4, R5 - V total & V total adj.
00030A 1026 18 A          FCB     $18,$19  R6, R7 - V displayed $ VS pos.
00031A 1028 00 A          FCB     $00,$0A  R8, R9 - Interlace & Max scan line
00032A 102A 00 A          FCB     $00,$0B  R10,R11- Cursor start & end
00033A 102C 0080 A        FDB     $0080    R12,R13- Start Address
00034A 102E 0080 A        FDB     $0080    R14,R15- Cursor Address
00035          END
TOTAL ERRORS 00000--00000

CRTC1  0004  CRTCAD 9000  CRTCRG 9001  CRTTAB 1020

```